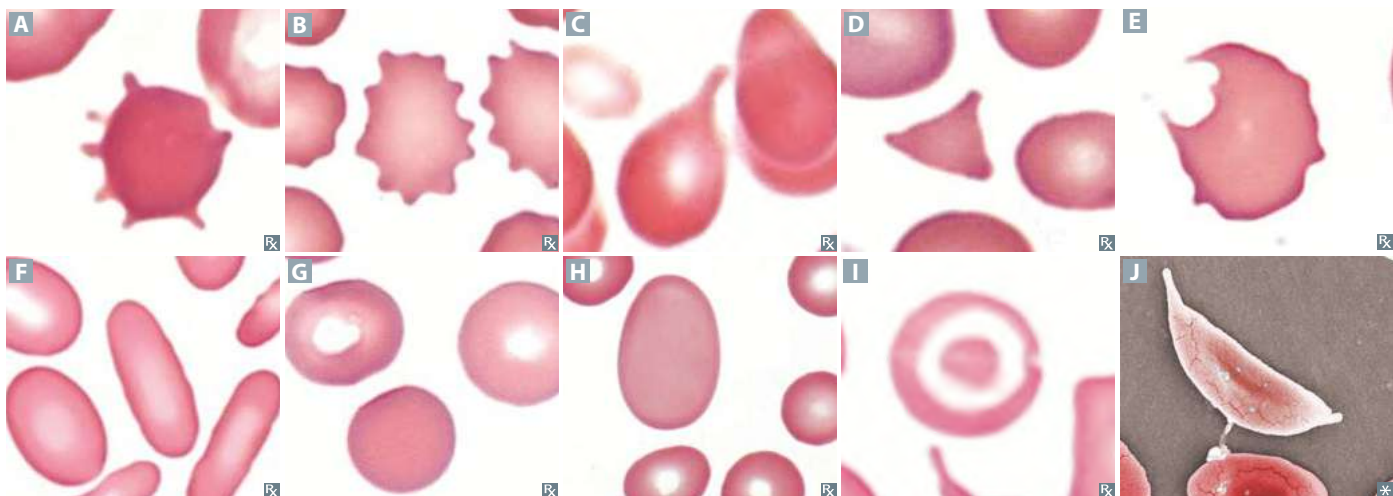


► HEMATOLOGY AND ONCOLOGY—PATHOLOGY

RBC morphology

TYPE	ASSOCIATED PATHOLOGY	NOTES
Acanthocytes A ("spur cells")	Liver disease, abetalipoproteinemia, vitamin E deficiency	Projections of varying size at irregular intervals (a canthocytes are a symmetric)
Echinocytes B ("burr cells")	Liver disease, ESRD, pyruvate kinase deficiency	Smaller and more uniform projections than acanthocytes (e chinocytes are e ven)
Dacryocytes C ("teardrop cells")	Bone marrow infiltration (eg, myelofibrosis)	RBC "sheds a tear " because it's mechanically squeezed out of its home in the bone marrow
Schistocytes D ("helmet" cells)	Microangiopathic hemolytic anemia (eg, DIC, TTP/HUS, HELLP syndrome), mechanical hemolysis (eg, heart valve prosthesis)	Fragmented RBCs
Degmacytes E ("bite cells")	G6PD deficiency	Due to removal of Heinz bodies by splenic macrophages (they " deg " them out of/ bite them off of RBCs)
Elliptocytes F	Hereditary elliptocytosis	Caused by mutation in genes encoding RBC membrane proteins (eg, spectrin)
Spherocytes G	Hereditary spherocytosis, autoimmune hemolytic anemia	Small, spherical cells without central pallor ↓ surface area-to-volume ratio
Macro-ovalocytes H	Megaloblastic anemia (also hypersegmented PMNs)	
Target cells I	H bC disease, A splenia, L iver disease, T halassemia	" HALT ," said the hunter to his target ↑ surface area-to-volume ratio
Sickle cells J	Sickle cell anemia	Sickling occurs with low O ₂ conditions (eg, high altitude, acidosis), high HbS concentration (ie, dehydration)



RBC inclusions

TYPE	ASSOCIATED PATHOLOGY	NOTES
Bone marrow		
Iron granules A	Sideroblastic anemias (eg, lead poisoning, myelodysplastic syndromes, chronic alcohol overuse)	Perinuclear mitochondria with excess iron (forming ring in ringed sideroblasts) Require Prussian blue stain to be visualized
Peripheral smear		
Howell-Jolly bodies B	Functional hyposplenism (eg, sickle cell disease), asplenia	Basophilic nuclear remnants (do not contain iron) Usually removed by splenic macrophages
Basophilic stippling C	Sideroblastic anemias, thalassemias	Basophilic ribosomal precipitates (do not contain iron)
Pappenheimer bodies	Sideroblastic anemias	Basophilic granules (contain iron) “Pappen-hammer” bodies
Heinz bodies D	G6PD deficiency	Denatured and precipitated hemoglobin (contain iron) Phagocytic removal of Heinz bodies → bite cells (take a bite of Heinz [ketchup]) Requires supravital stain (eg, crystal violet) to be visualized

A

B

C

D

↓

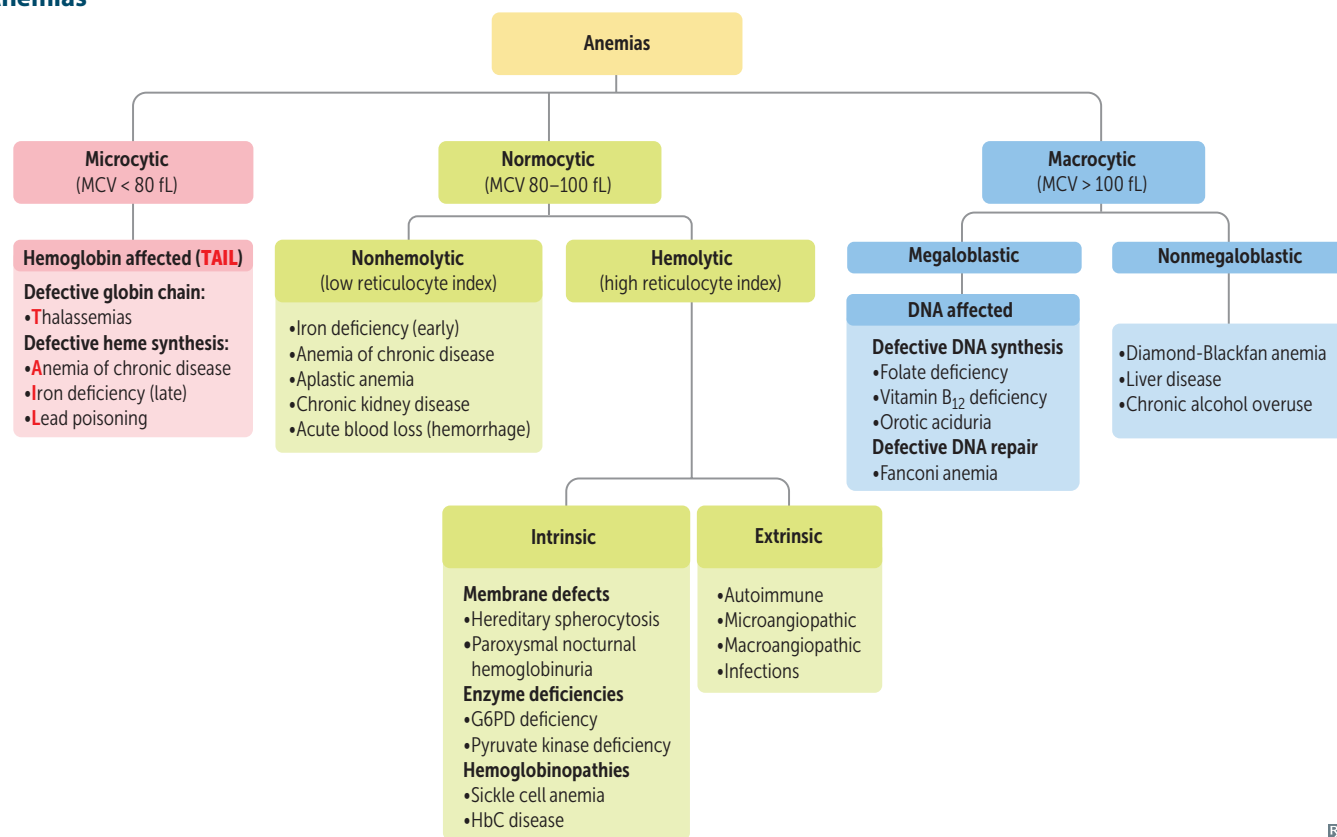
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Anemias



Reticulocyte production index

Also called corrected reticulocyte count. Used to correct falsely elevated reticulocyte count in anemia. Measures appropriate bone marrow response to anemic conditions (effective erythropoiesis). High RPI (> 3) indicates compensatory RBC production; low RPI (< 2) indicates inadequate response to correct anemia. Calculated as:

$$\text{RPI} = \% \text{ reticulocytes} \times \left(\frac{\text{actual Hct}}{\text{normal Hct}} \right) / \text{maturation time}$$

Mentzer index

Used to differentiate between thalassemia trait and iron deficiency anemia. An index of **less** than 13 suggests tha**les**semia trait. An **increased** index (> 13) suggests **iron** deficiency anemia.

$$\text{Mentzer index} = \frac{\text{MCV}}{\text{RBC count}}$$

Interpretation of iron studies

	Iron deficiency	Chronic disease	Hemochromatosis	Pregnancy/OCP use
Serum iron	↓	↓	↑	—
Transferrin or TIBC	↑	↓ ^a	↓	↑
Ferritin	↓	↑	↑	—
% transferrin saturation (serum iron/TIBC)	↓↓	—/↓	↑↑	↓

↑↓ = 1° disturbance.
Transferrin—**transports** iron in blood.
TIBC—indirectly measures transferrin.
Ferritin—1° iron storage protein of body.
^aEvolutionary reasoning—pathogens use circulating iron to thrive. The body has adapted a system in which iron is stored within the cells of the body and prevents pathogens from acquiring circulating iron.

Microcytic,**hypochromic anemias**

MCV < 80 fL.

Iron deficiency

↓ iron due to chronic bleeding (eg, GI loss, heavy menstrual bleeding), malnutrition, absorption disorders, GI surgery (eg, gastrectomy), or ↑ demand (eg, pregnancy) → ↓ final step in heme synthesis.

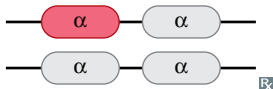
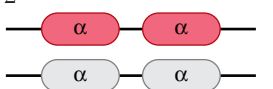
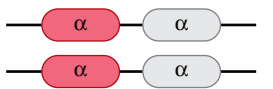
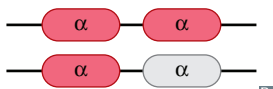
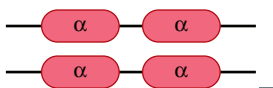
Labs: ↓ iron, ↑ TIBC, ↓ ferritin, ↑ free erythrocyte protoporphyrin, ↑ RDW, ↓ RI. Microcytosis and hypochromasia (↑ central pallor) **A**.

Symptoms: fatigue, conjunctival pallor **B**, restless leg syndrome, pica (persistent craving and compulsive eating of nonfood substances), spoon nails (koilonychia).

May manifest as glossitis, cheilosis, **Plummer-Vinson syndrome** (triad of iron deficiency anemia, esophageal webs, and dysphagia).

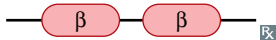
α-thalassemia

α-globin gene deletions on chromosome 16 → ↓ α-globin synthesis. May have *cis* deletion (deletions occur on same chromosome) or *trans* deletion (deletions occur on separate chromosomes). Normal is αα/αα. Often ↑ RBC count, in contrast to iron deficiency anemia. ↑ prevalence in people of Asian and African descent. Target cells **C** on peripheral smear.

# OF α-GLOBIN GENES DELETED	DISEASE	CLINICAL OUTCOME
1	α-thalassemia minima	No anemia (silent carrier)
		
2	α-thalassemia minor	Mild microcytic, hypochromic anemia
 Cis		
or		
 Trans		
3	Hemoglobin H disease (HbH); excess β-globin forms β ₄	Moderate to severe microcytic hypochromic anemia
		
4	Hemoglobin Barts disease; no α-globin, excess γ-globin forms γ ₄	Hydrops fetalis; incompatible with life
		

Microcytic, hypochromic anemias (continued) **β -thalassemia**

Point mutation in splice sites or Kozak consensus sequence (promoter) on chromosome 11 → ↓ β -globin synthesis (β^+) or absent β -globin synthesis (β^0). ↑ prevalence in Mediterranean populations.

# OF β -GLOBIN GENES MUTATED	DISEASE	CLINICAL OUTCOME
1 (β^+/β or β^0/β)	β -thalassemia minor	Mild microcytic anemia. ↑ HbA ₂ .
		
2 (β^+/β^+ or β^+/β^0)	β -thalassemia intermedia	Variable anemia, ranging from mild/asymptomatic to severe/transfusion-dependent.
2 (β^0/β^0)	β -thalassemia major (Cooley anemia)	Severe microcytic anemia with target cells and ↑ anisopoikilocytosis requiring blood transfusions (↑ risk of 2° hemochromatosis), marrow expansion (“crew cut” on skull x-ray D) → skeletal deformities, extramedullary hematopoiesis → HSM. ↑ risk of parvovirus B19-induced aplastic crisis. ↑ HbF and HbA ₂ , becomes symptomatic after 6 months when HbF declines (HbF is protective). Chronic hemolysis → pigmented gallstones.
		
1 (β^+/HbS or β^0/HbS)	Sickle cell β -thalassemia	Mild to moderate sickle cell disease depending on whether there is ↓ (β^+/HbS) or absent (β^0/HbS) β -globin synthesis.

Lead poisoning

Lead inhibits ferrochelatase and ALA dehydratase → ↓ heme synthesis and ↑ RBC protoporphyrin. Also inhibits rRNA degradation → RBCs retain aggregates of rRNA (basophilic stippling).

Symptoms of **L**LEEA**A**D poisoning:

- **L**ead **L**ines on gingivae (Burton lines) and on metaphyses of long bones **E** on x-ray.
- **E**ncephalopathy and **E**rythrocyte basophilic stippling.
- **A**bdominal colic and sideroblastic **A**nemia.
- **D**rops—wrist and foot drop.

Treatment: chelation with succimer, EDTA, dimercaprol.

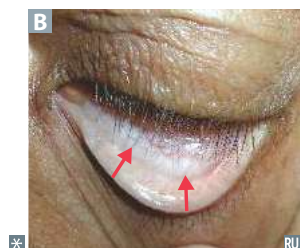
↑ exposure risk: children—chipped paint in old houses (built before 1978); adults—workplace (eg, batteries, ammunition).

Sideroblastic anemia

Causes: genetic (eg, X-linked defect in ALA synthase gene), acquired (myelodysplastic syndromes), and reversible (alcohol is most common; also lead poisoning, vitamin B₆ deficiency, copper deficiency, drugs [eg, isoniazid, linezolid]).

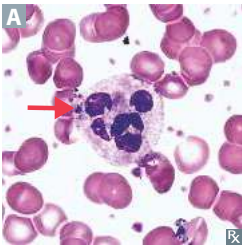
Lab findings: ↑ iron, normal/↓ TIBC, ↑ ferritin. Ringed sideroblasts (with iron-laden, Prussian blue–stained mitochondria) seen in bone marrow. Peripheral blood smear: basophilic stippling of RBCs. Some acquired variants may be normocytic or macrocytic.

Treatment: pyridoxine (B₆, cofactor for ALA synthase).

A

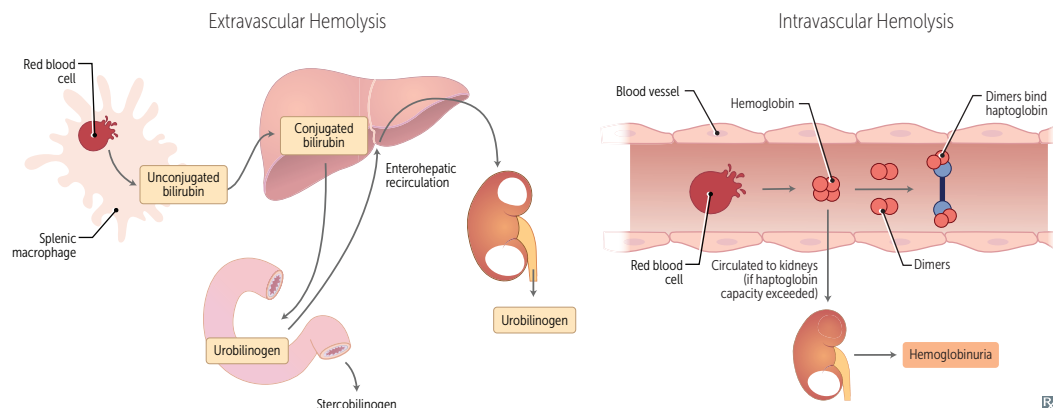
Macrocytic anemias

MCV > 100 fL.

	DESCRIPTION	FINDINGS
Megaloblastic anemia 	Impaired DNA synthesis → maturation of nucleus of precursor cells in bone marrow delayed relative to maturation of cytoplasm. Causes: vitamin B₁₂ deficiency, folate deficiency, medications (eg, hydroxyurea, phenytoin, methotrexate, sulfa drugs).	RBC macrocytosis, hypersegmented neutrophils (arrow in A), glossitis.
Folate deficiency	Causes: malnutrition (eg, chronic alcohol overuse), malabsorption, drugs (eg, methotrexate, trimethoprim, phenytoin), ↑ requirement (eg, hemolytic anemia, pregnancy).	↑ homocysteine, normal methylmalonic acid. No neurologic symptoms (vs B₁₂ deficiency).
Vitamin B₁₂ (cobalamin) deficiency	Causes: pernicious anemia, malabsorption (eg, Crohn disease), pancreatic insufficiency, gastrectomy, insufficient intake (eg, veganism), <i>Diphyllobothrium latum</i> (fish tapeworm).	↑ homocysteine, ↑ methylmalonic acid. Neurologic symptoms: reversible dementia, subacute combined degeneration (due to involvement of B₁₂ in fatty acid pathways and myelin synthesis): spinocerebellar tract, lateral corticospinal tract, dorsal column dysfunction. Folate supplementation in vitamin B₁₂ deficiency can correct the anemia, but worsens neurologic symptoms. Historically diagnosed with the Schilling test, a test that determines if the cause is dietary insufficiency vs malabsorption. Anemia 2° to insufficient intake may take several years to develop due to liver's ability to store B₁₂ (vs folate deficiency, which takes weeks to months).
Orotic aciduria	Inability to convert orotic acid to UMP (de novo pyrimidine synthesis pathway) because of defect in UMP synthase. Autosomal recessive. Presents in children as failure to thrive, developmental delay, and megaloblastic anemia refractory to folate and B₁₂. No hyperammonemia (vs ornithine transcarbamylase deficiency—↑ orotic acid with hyperammonemia).	Orotic acid in urine. Treatment: uridine monophosphate or uridine triacetate to bypass mutated enzyme.
Nonmegaloblastic anemia	Macrocytic anemia in which DNA synthesis is normal. Causes: chronic alcohol overuse, liver disease.	RBC macrocytosis without hypersegmented neutrophils.
Diamond-Blackfan anemia	A congenital form of pure red cell aplasia (vs Fanconi anemia, which causes pancytopenia). Rapid-onset anemia within 1st year of life due to intrinsic defect in erythroid progenitor cells.	↑ % HbF (but ↓ total Hb). Short stature, craniofacial abnormalities, and upper extremity malformations (triphalangeal thumbs) in up to 50% of cases. A pure Diamond causes pure red cell aplasia.

Normocytic, normochromic anemias

Normocytic, normochromic anemias are classified as nonhemolytic or hemolytic. The hemolytic anemias are further classified according to the cause of the hemolysis (intrinsic vs extrinsic to the RBC) and by the location of hemolysis (intravascular vs extravascular). Hemolysis can lead to ↑ in LDH, reticulocytes, unconjugated bilirubin, pigmented gallstones, and urobilinogen in urine.



Intravascular hemolysis

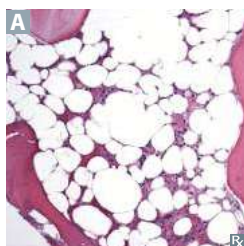
Findings: ↓ haptoglobin, ↑ schistocytes on blood smear. Characteristic hemoglobinuria, hemosiderinuria, and urobilinogen in urine. Notable causes are mechanical hemolysis (eg, prosthetic valve), paroxysmal nocturnal hemoglobinuria, microangiopathic hemolytic anemias.

Extravascular hemolysis

Mechanism: macrophages in spleen clear RBCs. Findings: splenomegaly, spherocytes in peripheral smear (most commonly due to hereditary spherocytosis and autoimmune hemolytic anemia), no hemoglobinuria/hemosiderinuria. Can present with urobilinogen in urine.

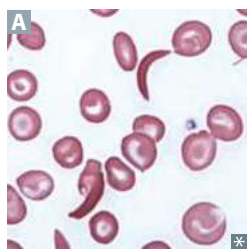
Nonhemolytic, normocytic anemias

	DESCRIPTION	FINDINGS
Anemia of chronic disease	Inflammation (eg, ↑ IL-6) → ↑ hepcidin (released by liver, binds ferroportin on intestinal mucosal cells and macrophages, thus inhibiting iron transport) → ↓ release of iron from macrophages and ↓ iron absorption from gut. Associated with conditions such as chronic infections, neoplastic disorders, chronic kidney disease, and autoimmune diseases (eg, SLE, rheumatoid arthritis).	↓ iron, ↓ TIBC, ↑ ferritin. Normocytic, but can become microcytic. Treatment: address underlying cause of inflammation, judicious use of blood transfusion, consider erythropoiesis-stimulating agents such as EPO (eg, in chronic kidney disease).
Aplastic anemia	Failure or destruction of hematopoietic stem cells. Causes (reducing volume from inside diaphysis): ▪ Radiation ▪ Viral agents (eg, EBV, HIV, hepatitis viruses) ▪ Fanconi anemia (autosomal recessive DNA repair defect → bone marrow failure); normocytosis or macrocytosis on CBC. Common associated findings include short stature, café-au-lait spots, thumb/radial defects, predisposition to malignancy. ▪ Idiopathic (immune mediated, 1° stem cell defect); may follow acute hepatitis ▪ Drugs (eg, benzene, chloramphenicol, alkylating agents, antimetabolites)	↓ reticulocyte count, ↑ EPO. Pancytopenia characterized by anemia, leukopenia, and thrombocytopenia (vs aplastic crisis, which causes anemia only). Normal cell morphology, but hypocellular bone marrow with fatty infiltration A (↑ adipose tissue in bone marrow in aplastic anemia). Symptoms: fatigue, malaise, pallor, purpura, mucosal bleeding, petechiae, infection. Treatment: withdrawal of offending agent, immunosuppressive regimens (eg, antithymocyte globulin, cyclosporine), bone marrow allograft, RBC/platelet transfusion, bone marrow stimulation (eg, GM-CSF).

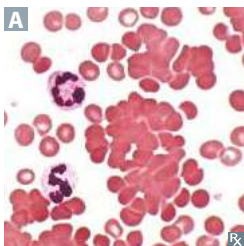


Intrinsic hemolytic anemias

	DESCRIPTION	FINDINGS
Hereditary spherocytosis	Primarily autosomal dominant. Due to defect in proteins interacting with RBC membrane skeleton and plasma membrane (eg, ankyrin, band 3, protein 4.2, spectrin). Small, round RBCs with no central pallor. ↓ surface area/dehydration → ↑ MCHC → premature removal by spleen (extravascular hemolysis).	Splenomegaly, pigmented gallstones, aplastic crisis (parvovirus B19 infection). Labs: ↓ mean fluorescence of RBCs in eosin 5-maleimide (EMA) binding test, ↑ fragility in osmotic fragility test (RBC hemolysis with exposure to hypotonic solution). Normal to ↓ MCV with abundance of RBCs. Treatment: splenectomy.
Paroxysmal nocturnal hemoglobinuria	Hematopoietic stem cell mutation → ↑ complement-mediated intravascular hemolysis, especially at night. Acquired <i>PIGA</i> mutation → impaired GPI anchor synthesis for decay-accelerating factor (DAF/CD55) and membrane inhibitor of reactive lysis (MIRL/CD59), which protect RBC membrane from complement.	Triad: Coombs ⊖ hemolytic anemia (mainly intravascular), pancytopenia, venous thrombosis (eg, Budd-Chiari syndrome). Pink/red urine in morning. Associated with aplastic anemia, acute leukemias. Labs: CD55/59 ⊖ RBCs on flow cytometry. Treatment: eculizumab (targets terminal complement protein C5).
G6PD deficiency	X-linked recessive. G6PD defect → ↓ NADPH → ↓ reduced glutathione → ↑ RBC susceptibility to oxidative stress (eg, sulfa drugs, antimalarials, fava beans) → hemolysis. Causes extravascular and intravascular hemolysis.	Back pain, hemoglobinuria a few days after oxidant stress . Labs: ↓ G6PD activity (may be falsely normal during acute hemolysis), blood smear shows RBCs with Heinz bodies and bite cells. “ Stress makes me eat bites of fava beans with Heinz ketchup.”
Pyruvate kinase deficiency	Autosomal recessive. Pyruvate kinase defect → ↓ ATP → rigid RBCs → extravascular hemolysis. Increases levels of 2,3-BPG → ↓ hemoglobin affinity for O ₂ .	Hemolytic anemia in a newborn. Labs: blood smear shows burr cells.
Sickle cell anemia	Point mutation in β-globin gene → single amino acid substitution (glutamic acid → valine) alters hydrophobic region on β-globin chain → aggregation of hemoglobin. Causes extravascular and intravascular hemolysis. Pathogenesis: ↓ O ₂ , ↑ altitude, or acidosis precipitates sickling (deoxygenated HbS polymerizes) → vaso-occlusive disease. Newborns are initially asymptomatic because of ↑ HbF and ↓ HbS. Heterozygotes (sickle cell trait) have resistance to malaria. Sickle cells are crescent-shaped RBCs A . “Crew cut” on skull x-ray due to marrow expansion from ↑ erythropoiesis (also seen in thalassemias).	Complications: ▪ Aplastic crisis (transient arrest of erythropoiesis due to parvovirus B19). ▪ Autosplenectomy (Howell-Jolly bodies) → ↑ risk of infection by encapsulated organisms (eg, <i>Salmonella</i> osteomyelitis). ▪ Splenic infarct/sequestration crisis. ▪ Painful vaso-occlusive crises: dactylitis (painful swelling of hands/feet), priapism, acute chest syndrome (respiratory distress, new pulmonary infiltrates on CXR, common cause of death), avascular necrosis, stroke. ▪ Sickling in renal medulla (↓ PO₂) → renal papillary necrosis → hematuria (also seen in sickle cell trait). Hb electrophoresis: ↓↓ HbA, ↑ HbF, ↑↑ HbS. Treatment: hydroxyurea (↑ HbF), hydration.
HbC disease	Glutamic acid→to-lycine (lysine) mutation in β-globin. Causes extravascular hemolysis.	HbSC (1 of each mutant gene) milder than HbSS. Blood smear in homozygotes: hemoglobin crystals inside RBCs, target cells.



Extrinsic hemolytic anemias

	DESCRIPTION	FINDINGS
Autoimmune hemolytic anemia 	A normocytic anemia that is usually idiopathic and Coombs ⊕. Two types: ▪ Warm AIHA—chronic anemia in which primarily IgG causes extravascular >>> intravascular hemolysis. Seen in SLE and CLL and with certain drugs (eg, β-lactams, α-methyl dopa). “Warm weather is Good.” ▪ Cold AIHA—acute anemia in which primarily IgM + complement cause RBC agglutination and extravascular >>> intravascular hemolysis upon exposure to cold → painful, blue fingers and toes. Seen in CLL, <i>Mycoplasma pneumoniae</i> infections, infectious mononucleosis.	Spherocytes and agglutinated RBCs A on peripheral blood smear. Warm AIHA treatment: steroids, rituximab, splenectomy (if refractory). Cold AIHA treatment: cold avoidance, rituximab.
Drug-induced hemolytic anemia	Most commonly due to antibody-mediated immune destruction of RBCs or oxidant injury via free radical damage (may be exacerbated in G6PD deficiency). Common causes include antibiotics (eg, penicillins, cephalosporins), NSAIDs, immunotherapy, chemotherapy.	Spherocytes suggest immune hemolysis. Bite cells suggest oxidative hemolysis. Can cause both extravascular and intravascular hemolysis.
Microangiopathic hemolytic anemia	RBCs are damaged when passing through obstructed or narrowed vessels. Causes intravascular hemolysis. Seen in DIC, TTP/HUS, SLE, HELLP syndrome, hypertensive emergency.	Schistocytes (eg, “helmet cells”) are seen on peripheral blood smear due to mechanical destruction (<i>schisto</i> = to split) of RBCs.
Macroangiopathic hemolytic anemia	Prosthetic heart valves and aortic stenosis may also cause hemolytic anemia 2° to mechanical destruction of RBCs.	Schistocytes on peripheral blood smear.
Hemolytic anemia due to infection	↑ destruction of RBCs (eg, malaria, <i>Babesia</i>).	

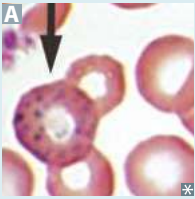
Leukopenias

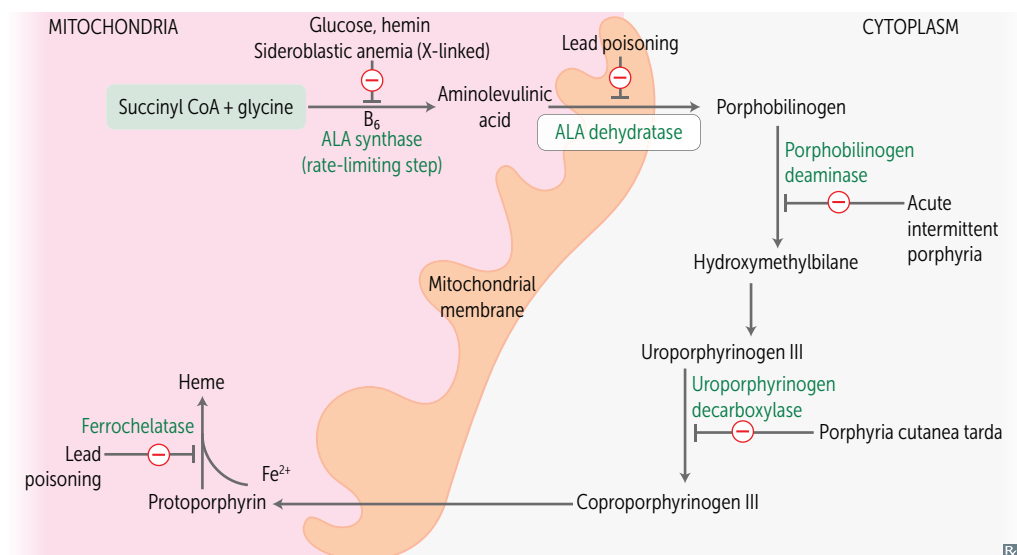
CELL TYPE	CELL COUNT	CAUSES
Neutropenia	Absolute neutrophil count < 1500 cells/mm³ Severe infections typical when < 500 cells/mm³	Sepsis/postinfection, drugs (including chemotherapy), aplastic anemia, autoimmunity (eg, SLE), radiation, congenital
Lymphopenia	Absolute lymphocyte count < 1500 cells/mm³ (< 3000 cells/mm³ in children)	HIV, DiGeorge syndrome, SCID, SLE, glucocorticoids^a, radiation, sepsis, postoperative
Eosinopenia	Absolute eosinophil count < 30 cells/mm³	Cushing syndrome, glucocorticoids^a

^aGlucocorticoids cause neutrophilia, despite causing eosinopenia and lymphopenia. Glucocorticoids ↓ activation of neutrophil adhesion molecules, impairing migration out of the vasculature to sites of inflammation. In contrast, glucocorticoids sequester eosinophils in lymph nodes and cause apoptosis of lymphocytes.

Heme synthesis, porphyrias, and lead poisoning

The porphyrias are hereditary or acquired conditions of defective heme synthesis that lead to the accumulation of heme precursors. Lead inhibits specific enzymes needed in heme synthesis, leading to a similar condition.

CONDITION	AFFECTED ENZYME	ACCUMULATED SUBSTRATE	PRESENTING SYMPTOMS
Lead poisoning 	Ferrochelatase and ALA dehydratase	Protoporphyrin, ALA (blood)	Microcytic anemia (basophilic stippling in peripheral smear A), ringed sideroblasts in bone marrow), GI and kidney disease.
Acute intermittent porphyria	Porphobilinogen deaminase (autosomal dominant mutation)	Porphobilinogen, ALA	Symptoms (5 P's): ▪ Painful abdomen ▪ Port wine-colored Pee ▪ Polyneuropathy ▪ Psychological disturbances ▪ Precipitated by factors that ↑ ALA synthase (eg, drugs [CYP450 inducers], alcohol, starvation) Treatment: hemin and glucose.
Porphyria cutanea tarda 	Uroporphyrinogen decarboxylase	Uroporphyrin (tea-colored urine)	Blistering cutaneous p hotosensitivity and hyperpigmentation B . Most common porphyria. Exacerbated with alcohol consumption. Causes: familial, hepatitis C . Treatment: phlebotomy, sun avoidance, antimalarials (eg, hydroxychloroquine).



Iron poisoning

Occurs via free radical formation and membrane lipid peroxidation → cell death.

	Acute	Chronic
FINDINGS	↑ mortality rate associated with accidental ingestion by children (adult iron tablets may look like candy).	Seen in patients with 1° (hereditary) or 2° (eg, chronic blood transfusions for thalassemia or sickle cell disease) hemochromatosis.
SYMPTOMS/SIGNS	Abdominal pain, vomiting, GI bleeding. Radiopaque pill seen on x-ray. May progress to anion gap metabolic acidosis and multiorgan failure. Leads to scarring with GI obstruction.	Arthropathy, cirrhosis, cardiomyopathy, diabetes mellitus and skin pigmentation (“bronze diabetes”), hypogonadism.
TREATMENT	Chelation (eg, deferoxamine, deferasirox), gastric lavage.	Chelation, regular therapeutic phlebotomy (unless contraindicated, eg, anemia)

Coagulation disorders

PT—tests function of common and extrinsic pathway (factors I, II, V, VII, and X). Defect → ↑ **PT** (Play **Tennis** **outside** [**extrinsic** pathway]).

INR (international normalized ratio) = patient PT/control PT. 1 = normal, > 1 = prolonged. Most common test used to follow patients on warfarin, which prolongs INR.

PTT—tests function of common and **intrinsic** pathway (all factors except VII and XIII). Defect → ↑ **PTT** (Play **Table Tennis** **inside**).

TT—measures the rate of conversion of fibrinogen → fibrin. Prolonged by anticoagulants, hypofibrinogenemia, DIC, liver disease.

Coagulation disorders can be due to clotting factor deficiencies or acquired factor inhibitors (most commonly against factor VIII). Diagnosed with a mixing study, in which normal plasma is added to patient’s plasma. Clotting factor deficiencies should correct (the PT or PTT returns to within the appropriate normal range), whereas factor inhibitors will not correct.

DISORDER	PT	PTT	MECHANISM AND COMMENTS
Hemophilia A, B, or C 	—	↑	Intrinsic pathway coagulation defect (↑ PTT). ▪ A: deficiency of factor VIII; X-linked recessive. Pronounce “hemophilia Ate (eight).” ▪ B: deficiency of factor IX; X-linked recessive. ▪ C: deficiency of factor XI; autosomal recessive. Hemorrhage in hemophilia—hemarthroses (bleeding into joints, eg, knee A), easy bruising, bleeding after trauma or surgery (eg, dental procedures). Treatment: desmopressin, factor VIII concentrate, emicizumab (A); factor IX concentrate (B); factor XI concentrate (C).
Vitamin K deficiency	↑	↑	General coagulation defect. Bleeding time normal. ↓ activity of factors II, VII, IX, X, protein C, protein S.

Platelet disorders

All platelet disorders have ↑ bleeding time (BT), mucous membrane bleeding, and microhemorrhages (eg, petechiae, epistaxis). Platelet count (PC) is usually low, but may be normal in qualitative disorders.

DISORDER	PC	BT	NOTES
Bernard-Soulier syndrome	−/↓	↑	Autosomal recessive defect in adhesion. ↓ GpIb → ↓ platelet-to-vWF adhesion. Labs: ↓ platelet aggregation, B ig platelets.
Glanzmann thrombasthenia	−	↑	Autosomal recessive defect in aggregation. ↓ GpIIb/IIIa (↓ integrin $\alpha_{IIb}\beta_3$) → ↓ platelet-to-platelet aggregation and defective platelet plug formation. Labs: blood smear shows no platelet clumping.
Immune thrombocytopenia	↓	↑	Destruction of platelets in spleen. Most commonly due to anti-GpIIb/IIIa antibodies → splenic macrophages phagocytose platelets. May be idiopathic or 2° to autoimmune disorders (eg, SLE), viral illness (eg, HIV, HCV), malignancy (eg, CLL), or drug reactions. Labs: ↑ megakaryocytes on bone marrow biopsy, ↓ platelet count. Treatment: glucocorticoids, IVIG, rituximab, TPO receptor agonists (eg, eltrombopag, romiplostim), or splenectomy for refractory ITP.
Uremic platelet dysfunction	−	↑	In patients with renal failure, uremic toxins accumulate and interfere with platelet adhesion and aggregation.

Thrombotic microangiopathies

Disorders overlap significantly in symptomatology. May resemble DIC, but do not exhibit lab findings of a consumptive coagulopathy (eg, ↑ PT, ↑ PTT, ↓ fibrinogen), as etiology does not involve widespread clotting factor activation.

	Thrombotic thrombocytopenic purpura	Hemolytic-uremic syndrome
EPIDEMIOLOGY	Typically females	Typically children
PATHOPHYSIOLOGY	Inhibition or deficiency of ADAMTS13 (a vWF metalloprotease) → ↓ degradation of vWF multimers → ↑ large vWF multimers → ↑ platelet adhesion and aggregation (microthrombi formation)	Predominately caused by Shiga toxin–producing <i>Escherichia coli</i> (STEC) infection (serotype O157:H7), which causes profound endothelial dysfunction.
PRESENTATION	Triad of thrombocytopenia (↓ platelets), microangiopathic hemolytic anemia (↓ Hb, schistocytes, ↑ LDH), acute kidney injury (↑ Cr)	
DIFFERENTIATING SYMPTOMS	Triad + fever + neurologic symptoms	Triad + bloody diarrhea
LABS	Normal PT and PTT helps distinguish TTP and HUS (coagulation pathway is not activated) from DIC (coagulation pathway is activated)	
TREATMENT	Plasma exchange, glucocorticoids, rituximab	Supportive care

Mixed platelet and coagulation disorders

DISORDER	PC	BT	PT	PTT	NOTES
von Willebrand disease	—	↑	—	—/↑	Intrinsic pathway coagulation defect: ↓ quantity/function of vWF → ↑ PTT (vWF carries/protects factor VIII). Defect in platelet plug formation: ↓ vWF → defect in platelet-to-vWF adhesion. Most are autosomal dominant. Mild but most common inherited bleeding disorder. Commonly presents with menorrhagia or epistaxis. Treatment: vWF concentrates, desmopressin (releases vWF stored in endothelium).
Disseminated intravascular coagulation	↓	↑	↑	↑	Widespread clotting factor activation → thromboembolic state with excessive clotting factor consumption → ↑ thromboses, ↑ hemorrhages (eg, blood oozing from puncture sites). May be acute (life-threatening) or chronic (if clotting factor production can compensate for consumption). Causes: heat Stroke, Snake bites, Sepsis (gram $\ominus$), Trauma, Obstetric complications, acute Pancreatitis, malignancy, nephrotic syndrome, transfusion (SSSTOP making new thrombi). Labs: schistocytes, ↑ fibrin degradation products (D-dimers), ↓ fibrinogen, ↓ factors V and VIII.

Hereditary thrombophilias Autosomal dominant disorders resulting in hypercoagulable state (↑ tendency to develop thrombosis).

DISEASE	DESCRIPTION
Antithrombin deficiency	Has no direct effect on the PT, PTT, or thrombin time but diminishes the increase in PTT following standard heparin dosing. Can also be acquired: renal failure/nephrotic syndrome → antithrombin loss in urine → ↓ inhibition of factors IIa and Xa.
Factor V Leiden	Production of mutant factor V (guanine → adenine DNA point mutation → Arg506Gln mutation near the cleavage site) that is resistant to degradation by activated protein C. Complications include DVT, cerebral vein thrombosis, recurrent pregnancy loss.
Protein C or S deficiency	↓ ability to inactivate factors Va and VIIIa. ↑ risk of warfarin-induced skin necrosis. Together, protein C Cancels, and protein S Stops, coagulation.
Prothrombin G20210A mutation	Point mutation in 3' untranslated region → ↑ production of prothrombin → ↑ plasma levels and venous clots.

Blood transfusion therapy

COMPONENT	DOSAGE EFFECT	CLINICAL USE
Packed RBCs	↑ Hb and O ₂ binding (carrying) capacity, ↑ hemoglobin ~1 g/dL per unit, ↑ hematocrit ~3% per unit	Acute blood loss, severe anemia
Platelets	↑ platelet count ~30,000/microL per unit (↑ ~5000/mm ³ /unit)	Stop significant bleeding (thrombocytopenia, qualitative platelet defects)
Fresh frozen plasma/ prothrombin complex concentrate	↑ coagulation factor levels; FFP contains all coagulation factors and plasma proteins; PCC generally contains factors II, VII, IX, and X, as well as protein C and S	Cirrhosis, immediate anticoagulation reversal
Cryoprecipitate	Contains fibrinogen, factor VIII, factor XIII, vWF, and fibronectin	Coagulation factor deficiencies involving fibrinogen and factor VIII
Albumin	↑ intravascular volume and oncotic pressure	Post-paracentesis, therapeutic plasmapheresis

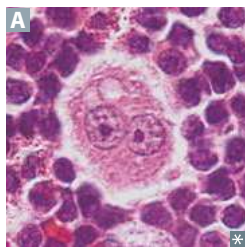
Blood transfusion risks include infection transmission (low), transfusion reactions, transfusion-associated circulatory overload (TACO; volume overload → pulmonary edema, hypertension), transfusion-related acute lung injury (TRALI; hypoxia and inflammation → noncardiogenic pulmonary edema, hypotension), iron overload (may lead to 2° hemochromatosis), hypocalcemia (citrate is a Ca²⁺ chelator), and hyperkalemia (RBCs may lyse in old blood units).

Leukemia vs lymphoma

Leukemia	Lymphoid or myeloid neoplasm with widespread involvement of bone marrow. Tumor cells are usually found in peripheral blood.
Lymphoma	Discrete tumor mass arising from lymph nodes. Variable clinical presentation (eg, arising in atypical sites, leukemic presentation).

Hodgkin vs non-Hodgkin lymphoma

Hodgkin	Non-Hodgkin
Both may have constitutional (“B”) signs/symptoms: low-grade fever, night sweats, weight loss.	
Localized, single group of nodes with contiguous spread (stage is strongest predictor of prognosis). Better prognosis.	Multiple lymph nodes involved; extranodal involvement common; noncontiguous spread. Worse prognosis.
Characterized by Reed-Sternberg cells.	Majority involve B cells; rarely of T-cell lineage.
Bimodal distribution: young adults, > 55 years.	Can occur in children and adults.
Associated with EBV.	May be associated with autoimmune diseases and viral infections (eg, HIV, EBV, HTLV).

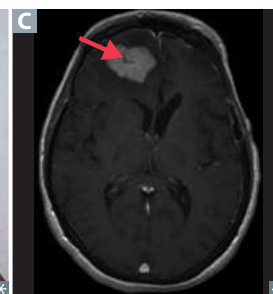
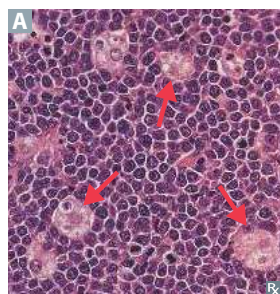
Hodgkin lymphoma

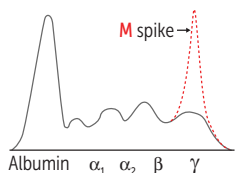
Contains Reed-Sternberg cells: distinctive tumor giant cells; bilobed nucleus with the **2** halves as mirror images (“owl eyes” **A**). RS cells are CD**15**⁺ and CD**30**⁺ B-cell origin. **2** owl eyes × **15** = **30**.

SUBTYPE	NOTES
Nodular sclerosis	Most common
Mixed cellularity	Eosinophilia; seen in immunocompromised patients
Lymphocyte rich	Best prognosis (the rich have better bank accounts)
Lymphocyte depleted	Worst prognosis (the poor have worse bank accounts); seen in immunocompromised patients

Non-Hodgkin lymphoma

TYPE	OCCURS IN	GENETICS	COMMENTS
Neoplasms of mature B cells			
Burkitt lymphoma	Adolescents or young adults “Bur kid ” lymphoma (more common in kids)	t(8;14)—translocation of <i>c-myc</i> (8) and heavy-chain Ig (14)	“ Starry sky ” appearance (StarBurst), sheets of lymphocytes with interspersed “tingible body” macrophages (arrows in A). Associated with EBV. Jaw lesion B in endemic form in Africa; pelvis or abdomen in sporadic form.
Diffuse large B-cell lymphoma	Usually older adults, but 20% in children	Mutations in <i>BCL-2</i> , <i>BCL-6</i>	Most common type of non-Hodgkin lymphoma in adults.
Follicular lymphoma	Adults	t(14;18)—translocation of heavy-chain Ig (14) and <i>BCL-2</i> (18)	Indolent course with painless “waxing and waning” lymphadenopathy. Bcl-2 normally inhibits apoptosis.
Mantle cell lymphoma	Adult males >> adult females	t(11;14)—translocation of cyclin D1 (11) and heavy-chain Ig (14), CD5+	Very aggressive, patients typically present with late-stage disease.
Marginal zone lymphoma	Adults	t(11;18)	Associated with chronic inflammation (eg, Sjögren syndrome, chronic gastritis [MALT lymphoma; may regress with <i>H pylori</i> eradication]).
Primary central nervous system lymphoma	Adults	EBV related; associated with HIV/AIDS	Considered an AIDS-defining illness. Variable presentation: confusion, memory loss, seizures. CNS mass (often single, ring-enhancing lesion on MRI) in immunocompromised patients C , needs to be distinguished from toxoplasmosis via CSF analysis or other lab tests.
Neoplasms of mature T cells			
Adult T-cell lymphoma	Adults	Caused by HTLV (associated with IV drug use)	Adults present with cutaneous lesions; common in Japan (T -cell in T okyo), West Africa, and the Caribbean. Lytic bone lesions, hypercalcemia.
Cutaneous T-cell lymphoma	Adults		Heterogenous group of T-cell neoplasms affecting the skin ± blood, lymph nodes, or viscera. Most common subtype is mycosis fungoides D characterized by erythematous patches favoring sun-protected areas that progress to plaques, then eventually tumors.



Plasma cell dyscrasias

Group of disorders characterized by proliferation of a single plasma cell clone, typically overproducing a monoclonal immunoglobulin (also called paraprotein). Seen in older adults. Screening with serum protein electrophoresis (**M** spike represents overproduction of **M**onoclonal Ig), serum immunofixation, and serum free light chain assay. Urine protein electrophoresis and immunofixation required to confirm urinary involvement (urine dipstick only detects albumin). Diagnostic confirmation with bone marrow biopsy.

Peripheral blood smear may show rouleaux formation **A** (RBCs stacked like poker chips).

Multiple myeloma

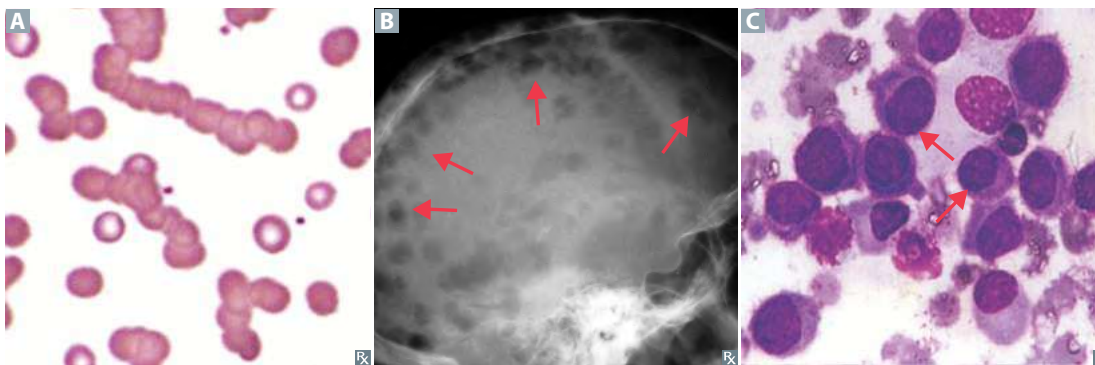
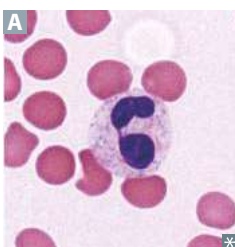
Overproduction of IgG (most common) > IgA > Ig light chains. Clinical features (**CRAB**): hyper**C**alcemia ($\uparrow$ cytokine secretion [eg, IL-1, TNF- α , RANK-L] by malignant plasma cells $\rightarrow$ $\uparrow$ osteoclast activity), **R**enal insufficiency, **A**nemia, **B**one lytic lesions (“punched out” on x-ray **B** $\rightarrow$ back pain, pathologic fractures). Complications: $\uparrow$ infection risk, 1° amyloidosis (AL). Urinalysis may show Ig light chains (Bence Jones proteinuria) with $\ominus$ urine dipstick. Bone marrow biopsy shows >10% monoclonal plasma cells with clock-face chromatin **C** and intracytoplasmic inclusions containing Ig.

Waldenström macroglobulinemia

Overproduction of IgM (**mac**roglobulinemia because Ig**M** is the **largest** Ig). Clinical features include anemia, constitutional (“B”) signs/symptoms, lymphadenopathy, hepatosplenomegaly, hyperviscosity (eg, headache, bleeding, blurry vision, ataxia), peripheral neuropathy. Funduscopy shows dilated, segmented, and tortuous retinal veins (sausage link appearance). Bone marrow biopsy shows >10% monoclonal B lymphocytes with plasma cell features (lymphoplasmacytic lymphoma) and intranuclear pseudoinclusions containing IgM.

Monoclonal gammopathy of undetermined significance

Overproduction of any Ig type (M spike <3 g/dL). Asymptomatic (no CRAB findings). 1%–2% risk per year of progressing to multiple myeloma. Bone marrow biopsy shows <10% monoclonal plasma cells.

**Myelodysplastic syndromes**

Stem cell disorders involving ineffective hematopoiesis $\rightarrow$ defects in cell maturation of nonlymphoid lineages. Bone marrow blasts < 20% (vs > 20% in AML). Caused by de novo mutations or environmental exposure (eg, radiation, benzene, chemotherapy). Risk of transformation to AML. More common in older adults.

Pseudo-Pelger-Huët anomaly—neutrophils with bilobed (“**duet**”) nuclei **A**. Associated with myelodysplastic syndromes or drugs (eg, immunosuppressants).

Leukemias

Unregulated growth and differentiation of WBCs in bone marrow → marrow failure → anemia (↓ RBCs), infections (↓ mature WBCs), and hemorrhage (↓ platelets). Usually presents with ↑ circulating WBCs (malignant leukocytes in blood), although some cases present with normal/↓ WBCs.

Leukemic cell infiltration of liver, spleen, lymph nodes, and skin (leukemia cutis) possible.

TYPE

NOTES

Lymphoid neoplasms**Acute lymphoblastic leukemia/lymphoma**

Most frequently occurs in children; less common in adults (worse prognosis). T-cell ALL can present as mediastinal mass (presenting as SVC-like syndrome). Associated with Down syndrome. Peripheral blood and bone marrow have ↑↑↑ lymphoblasts **A**.

TdT+ (marker of pre-T and pre-B cells), CD10+ (marker of pre-B cells).

Most responsive to therapy.

May spread to CNS and testes.

t(12;21) → better prognosis; t(9;22) (Philadelphia chromosome) → worse prognosis.

Chronic lymphocytic leukemia/small lymphocytic lymphoma

Age > 60 years. Most common adult leukemia. CD20+, CD23+, CD5+ B-cell neoplasm. Often asymptomatic, progresses slowly; smudge cells **B** in peripheral blood smear; autoimmune hemolytic anemia. **CLL** = **C**rushed **L**ittle **L**ymphocytes (smudge cells).

Richter transformation—CLL/SLL transformation into an aggressive lymphoma, most commonly diffuse large B-cell lymphoma (DLBCL).

Hairy cell leukemia

Adult males. Mature B-cell tumor. Cells have filamentous, hairlike projections (fuzzy appearing on LM **C**). Peripheral lymphadenopathy is uncommon.

Causes marrow fibrosis → dry tap on aspiration. Patients usually present with massive splenomegaly and pancytopenia.

Stains **TRAP** (**T**artrate-**R**esistant **A**cid **P**hosphatase) ⊕ (**TRAP**ped in a **hairy** situation). TRAP stain largely replaced with flow cytometry. Associated with *BRAF* mutations.

Treatment: purine analogs (cladribine, pentostatin).

Myeloid neoplasms**Acute myelogenous leukemia**

Median onset 65 years. Auer rods **D**; myeloperoxidase ⊕ cytoplasmic inclusions seen mostly in APL (formerly M3 AML); ↑↑↑ circulating myeloblasts on peripheral smear. May present with leukostasis (capillary occlusion by malignant, nondistensible cells → organ damage).

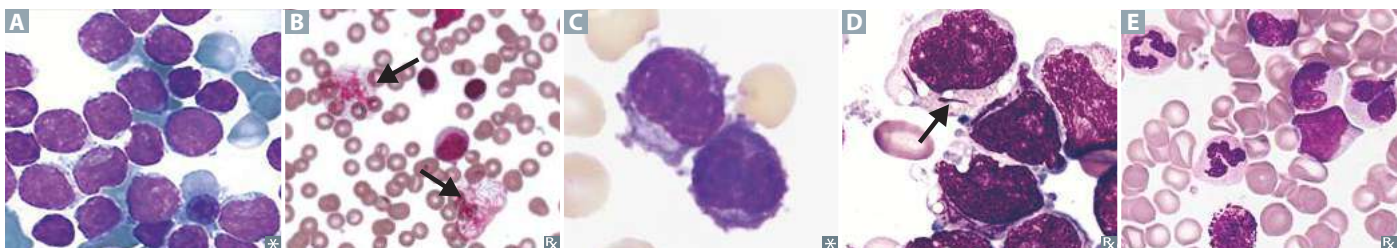
Risk factors: prior exposure to alkylating chemotherapy, radiation, benzene, myeloproliferative disorders, Down syndrome (typically acute megakaryoblastic leukemia [formerly M7 AML]).

APL: t(15;17), responds to all-*trans* retinoic acid (vitamin A) and arsenic trioxide, which induce differentiation of promyelocytes; DIC is a common presentation.

Chronic myelogenous leukemia

Peak incidence: 45–85 years; median age: 64 years. Defined by the Philadelphia chromosome (t[9;22], *BCR-ABL*) and myeloid stem cell proliferation. Presents with dysregulated production of mature and maturing granulocytes (eg, neutrophils, metamyelocytes, myelocytes, basophils **E**) and splenomegaly. May accelerate and transform to AML or ALL (“blast crisis”).

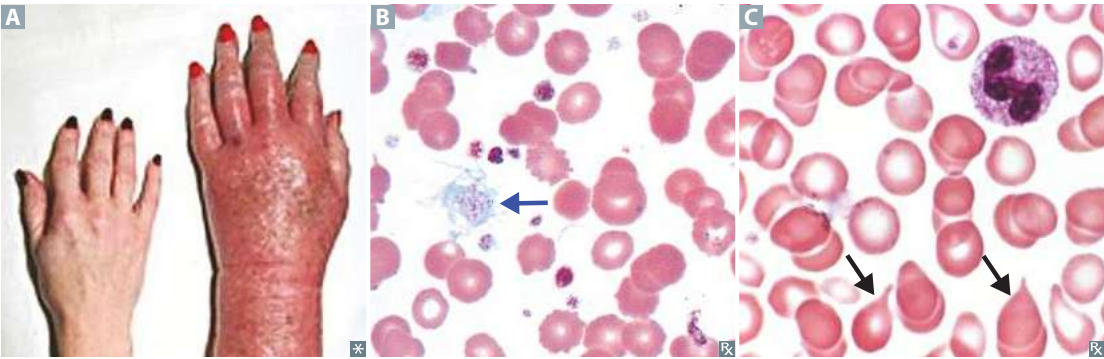
Responds to *BCR-ABL* tyrosine kinase inhibitors (eg, imatinib).



Myeloproliferative neoplasms Malignant hematopoietic neoplasms with varying impacts on WBCs and myeloid cell lines.

Polycythemia vera	Primary polycythemia. Disorder of ↑ RBCs, usually due to acquired <i>JAK2</i> mutation. May present as intense itching after shower (aquagenic pruritus). Rare but classic symptom is erythromelalgia (severe, burning pain and red-blue coloration) due to episodic blood clots in vessels of the extremities A . Associated with hyperviscosity and thrombosis (eg, PE, DVT, Budd-Chiari syndrome). ↓ EPO (vs 2° polycythemia, which presents with endogenous or artificially ↑ EPO). Treatment: phlebotomy, hydroxyurea, ruxolitinib (<i>JAK1/2</i> inhibitor).
Essential thrombocythemia	Characterized by massive proliferation of megakaryocytes and platelets. Symptoms include bleeding and thrombosis. Blood smear shows markedly increased number of platelets, which may be large or otherwise abnormally formed B . Erythromelalgia may occur.
Myelofibrosis	Atypical megakaryocyte hyperplasia → ↑ TGF-β secretion → ↑ fibroblast activity → obliteration of bone marrow with fibrosis. Associated with massive splenomegaly and “teardrop” RBCs C . “Bone marrow cries because it’s fibrosed and is a dry tap.”

	RBCs	WBCs	PLATELETS	PHILADELPHIA CHROMOSOME	<i>JAK2</i> MUTATIONS
Polycythemia vera	↑	↑	↑	⊖	⊕
Essential thrombocythemia	—	—	↑	⊖	⊕ (30–50%)
Myelofibrosis	↓	Variable	Variable	⊖	⊕ (30–50%)
CML	↓	↑	↑	⊕	⊖



Leukemoid reaction vs chronic myelogenous leukemia

	Leukemoid reaction	Chronic myelogenous leukemia
DEFINITION	Reactive neutrophilia > 50,000 cells/mm ³	Myeloproliferative neoplasm ⊕ for <i>BCR-ABL</i>
NEUTROPHIL MORPHOLOGY	Toxic granulation, Döhle bodies, cytoplasmic vacuoles	Pseudo-Pelger-Huët anomaly
LAP SCORE	↑	↓ (LAP enzyme ↓ in malignant neutrophils)
EOSINOPHILS AND BASOPHILS	Normal	↑

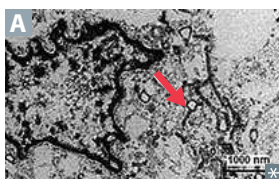
Polycythemia

	PLASMA VOLUME	RBC MASS	O ₂ SATURATION	EPO LEVELS	ASSOCIATIONS
Relative	↓	—	—	—	Dehydration, burns.
Appropriate absolute	—	↑	↓	↑	Lung disease, congenital heart disease, high altitude, obstructive sleep apnea.
Inappropriate absolute	—	↑	—	↑	Exogenous EPO (athlete misuse, also called “blood doping”), androgen supplementation. Inappropriate EPO secretion: malignancy (eg, RCC, HCC).
Polycythemia vera	↑	↑↑	—	↓	EPO ↓ in PCV due to negative feedback suppressing renal EPO production.

↑↓ = 1° disturbance

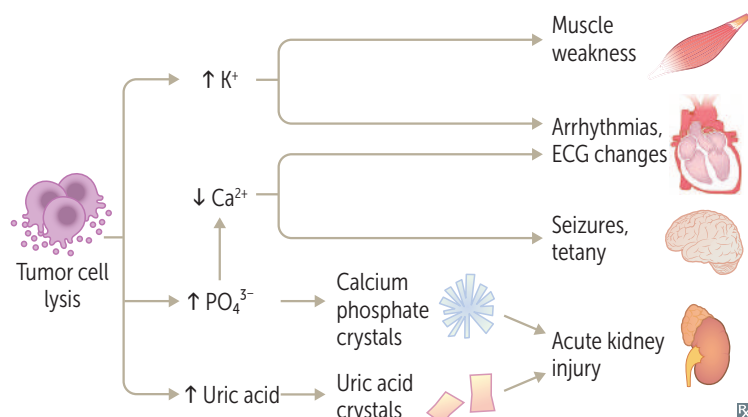
Chromosomal translocations

TRANSLOCATION	ASSOCIATED DISORDER	NOTES
t(8;14)	Burkitt (Burk-8) lymphoma (<i>c-myc</i> activation)	The Ig heavy chain genes on chromosome 14 are constitutively expressed. When other genes (eg, <i>c-myc</i> and <i>BCL-2</i>) are translocated next to this heavy chain gene region, they are overexpressed.
t(11;14)	Mantle cell lymphoma (cyclin D1 activation)	
t(11;18)	Marginal zone lymphoma	
t(14;18)	Follicular lymphoma (<i>BCL-2</i> activation)	
t(15;17)	APL (formerly M3 type of AML)	
t(9;22) (Philadelphia chromosome)	CML (<i>BCR-ABL</i> hybrid), ALL (less common); Philadelphia CreaML cheese	

Langerhans cell histiocytosis

Collective group of proliferative disorders of Langerhans cells (antigen-presenting cells normally found in the skin). Presents in a child as lytic bone lesions and skin rash or as recurrent otitis media with a mass involving the mastoid bone. Cells are functionally immature and do not effectively stimulate primary T cells via antigen presentation. Cells express S-100 and CD1a. Birbeck granules (“tennis rackets” or rod-shaped on EM) are characteristic **A**.

Tumor lysis syndrome



Oncologic emergency triggered by massive tumor cell lysis, seen most often with lymphomas/leukemias. Usually caused by treatment initiation, but can occur spontaneously with fast-growing cancers. Release of K^+ $\rightarrow$ hyperkalemia, release of PO_4^{3-} $\rightarrow$ hyperphosphatemia, hypocalcemia due to Ca^{2+} sequestration by PO_4^{3-} . $\uparrow$ nucleic acid breakdown $\rightarrow$ hyperuricemia $\rightarrow$ acute kidney injury. Prevention and treatment include aggressive hydration, allopurinol, rasburicase. Tumor cell breakdown causes $\uparrow K^+$, $\downarrow Ca^{2+}$, $\uparrow$ Uric acid, and $\uparrow PO_4^{3-}$ (Break the CUP).

► HEMATOLOGY AND ONCOLOGY—PHARMACOLOGY

Heparin

MECHANISM

Activates antithrombin, which $\downarrow$ action primarily of factors IIa (thrombin) and Xa. Short half-life.

CLINICAL USE

Immediate anticoagulation for pulmonary embolism (PE), acute coronary syndrome, MI, deep venous thrombosis (DVT). Used during pregnancy (does not cross placenta; low-molecular-weight heparin [eg, enoxaparin] preferred). Monitor PTT.

ADVERSE EFFECTS

Bleeding (reverse with protamine sulfate), heparin-induced thrombocytopenia (HIT), osteoporosis (with long-term use), drug-drug interactions, type 4 renal tubular acidosis.

- **HIT type 1**—mild (platelets $> 100,000/mm^3$), transient, nonimmunologic drop in platelet count that typically occurs within the first 2 days of heparin administration. Not clinically significant.
- **HIT type 2**—development of IgG antibodies against heparin-bound platelet factor 4 (PF4) that typically occurs 5–10 days after heparin administration. Antibody-heparin-PF4 complex binds and activates platelets $\rightarrow$ removal by splenic macrophages and thrombosis $\rightarrow$ $\downarrow\downarrow$ platelet count. Highest risk with unfractionated heparin. Treatment: discontinue heparin, start alternative anticoagulant (eg, argatroban). Fondaparinux safe to use (does not interact with PF4).

NOTES

Low-molecular-weight heparins (eg, enoxaparin, dalteparin) act mainly on factor Xa. Longer half-life. Fondaparinux acts only on factor Xa. Both are not easily reversible. Unfractionated heparin used in patients with renal insufficiency (low-molecular-weight heparins should be used with caution because they undergo renal clearance).